Diagnosing Lexical-Matching Limitations Across Domains: Experimental Results

Classical IR (Boolean, TF-IDF/VSM, BM25) on 8 BEIR datasets, with a vocabulary-overlap analysis, query-expansion and dense/hybrid comparisons. Every verdict below is produced by a decision rule fixed in code before inspecting the outcome (src/analysis.py, src/failures.py).

Summary of verdicts

The proposal's claims were turned into six testable hypotheses. Two are not supported (H1, H2), two are partially supported (H5, H6), and two are supported (H3, and H4 with caveats about label accuracy). The most consequential result is that **domain-level vocabulary specificity (jargon) does not explain where BM25 succeeds or fails**, whereas **query-level overlap does**, but in a way that is only partly specific to lexical retrieval and cannot be measured at query time.

#	Claim (from proposal)	Test	Verdict	Key evidence
H1	BM25 > TF-IDF/VSM > Boolean on every domain (Objective 2)	Paired randomization test on nDCG@10, Holm-corrected over 32 tests	**NOT SUPPORTED**	BM25 significantly beats TF-IDF on only 4/8 datasets; TF-IDF significantly beats BM25 on ArguAna (0.534 vs 0.402). TF-IDF > Boolean-OR on 6/8 (reversed on Touche). BM25 > Boolean on 8/8.
H2	Lexical retrieval degrades in domains of higher term specificity (Problem statement)	Spearman across 8 domains: measured jargon rate vs BM25 nDCG@10	**NOT SUPPORTED**	rho(corpus jargon, BM25) = 0.33 (p = 0.420), i.e. wrong sign and not significant. The proposal's specificity ordering of NFCorpus, SciFact and FiQA does not match the measured one: FiQA is the *least* jargon-heavy.
H3	Query-document vocabulary overlap explains BM25 effectiveness (Objective 3)	Partial Spearman per dataset + linear mixed model (random intercept per dataset)	**SUPPORTED**	Positive in 7/8 datasets (partial rho 0.37-0.73; Touche null). Mixed model: +0.210 nDCG per SD of overlap. **But** overlap also predicts the dense model (+0.106), so about half the effect is general query difficulty.
H4	Failures fall into identifiable modes that differ by domain (Objective 4)	Heuristic taxonomy, chi-square across domains, manual audit	**SUPPORTED**	Modes differ by domain (chi2 = 498, Cramer's V = 0.24, p < 0.001). Labels are heuristic: manual audit accuracy 78% (95% CI 64%-87%, n=45). Paraphrase/indirect relevance and term competition dominate; abbreviation is rare.
H5	Dense retrieval helps most where overlap is low (Objective 5)	Partial Spearman of (dense - BM25) gap vs overlap; mixed model	**PARTIALLY SUPPORTED**	Pooled effect is clear (mixed-model coef -0.104, p < 0.001) but significant in only 4/8 datasets for MiniLM; bge-small: 5/8. Not significant on FiQA, SciDocs, TREC-COVID or Touche.
H6	An overlap threshold tells when BM25 alone is defensible (Deliverable 4)	Leave-one-dataset-out threshold rule, balanced accuracy, AUC	**PARTIALLY SUPPORTED**	The learned threshold is stable (coverage about 0.60 in 7/8 folds) but transfers poorly: mean held-out balanced accuracy 0.56, only 3/8 folds >= 0.55; AUC 0.66. Overlap needs relevance labels, so it cannot route live queries.

What you may not want to hear

• **The central framing of the proposal (that lexical retrieval degrades with domain specificity) is not supported by this data.** The direction is, if anything, reversed: jargon-heavy collections (SciFact, TREC-COVID, NFCorpus) are where BM25 is relatively *strongest* against dense models (rho = -0.60 for jargon vs. MiniLM's gain, p = 0.120, n = 8). This matches the proposal's own Section 7 preprints, so the proposal currently argues both directions and must pick one.

• **The overlap-effectiveness correlation (Objective 3) is partly circular.** BM25 scores documents by shared terms, so a positive correlation is close to guaranteed. The informative contrast is that overlap predicts the dense model about half as strongly. Report the BM25-minus-dense difference, not the raw correlation, as the contribution.

• **A 'vocabulary-overlap threshold' (Deliverable 4) cannot be applied at query time.** Overlap is defined against the relevant documents, which are unknown for a new query. The threshold that was learned (coverage about 0.6) barely transfers to unseen domains (balanced accuracy 0.56).

• **The finance hypothesis borrowed from the 2026 preprint fails here.** On FiQA, even small dense models beat BM25 by 0.132 (MiniLM) and 0.154 (bge-small) nDCG@10, the largest gains among the original three domains. FiQA is opinion-style forum text, not ticker-heavy financial documents.

• **Domain-level conclusions rest on n = 8 points** (n = 3 in the original design, which could not support any correlation). They are underpowered; per-query analyses (about 14,000 queries) carry the inferential weight.

• **'Dense vs BM25' depends on which dense model is used.** On TREC-COVID, MiniLM loses to BM25 by 0.123 while bge-small wins by 0.151. Claims about 'semantic retrieval' in general should not be drawn from one model.

1. Setup

Data. The proposal's three BEIR datasets (NFCorpus, SciFact, FiQA-2018) plus five more for domain-level power: SciDocs (scientific citation prediction), ArguAna and Touche-2020 (argument retrieval), TREC-COVID (biomedical) and Quora (general-domain duplicate questions). Test splits; queries without a positive judgement are dropped; a document whose id equals the query id is removed from rankings (BEIR convention).

Lexical systems. Lucene-style BM25 (k1 = 0.9, b = 0.4, the Pyserini BEIR defaults), TF-IDF/VSM (SMART ltc, cosine), Boolean OR ranked by coordination level, and Boolean AND (set retrieval; random order within the set). Analysis: \w+ tokens, lower-case, Lucene stopword list, Porter stemmer. Title and text are indexed together ('flat'). Four preprocessing variants form the E2 ablation.

Expansion (E5). RM3 pseudo-relevance feedback (10 docs, 10 terms, original weight 0.5) and WordNet synonym expansion (up to 3 synonyms per term from the first two senses, weight 0.3).

Dense and hybrid (E6, added). all-MiniLM-L6-v2 and bge-small-en-v1.5, exact cosine search, 256 tokens, run on a Colab T4 GPU; hybrids by Reciprocal Rank Fusion (k = 60) and by convex combination of min-max-normalised scores (alpha = 0.5). CPU re-runs of MiniLM on five datasets agree with the GPU run to within 0.0004 mean nDCG@10.

Overlap features (E3). Primary, fixed in advance: *coverage* = share of the query's distinct (stemmed, stopped) terms present in a relevant document, averaged over that query's relevant documents. Robustness variants: IDF-weighted coverage, maximum coverage, Jaccard. Domain specificity is *measured* as the share of corpus and query tokens with general-English Zipf frequency < 3 (wordfreq), plus mean query IDF.

Statistics. Paired two-sided randomization tests (10,000 sign flips) with Holm correction; partial Spearman correlations controlling for number of query terms and log number of relevant documents, with 95% bootstrap CIs; linear mixed models with a random intercept per dataset; leave-one-dataset-out validation for the threshold. Seed 42 throughout.

Dataset	Domain	Docs	Test q.	Avg doc len	Avg q len	Rel/q	Corpus jargon	Query jargon	Mean q IDF	Mean coverage
NFCorpus	biomedical	3,633	323	245	3.4	38.2	0.121	0.267	4.33	0.22
SciFact	scientific	5,183	300	227	13.1	1.1	0.186	0.254	3.56	0.59
FiQA-2018	financial	57,638	648	142	11.2	2.6	0.034	0.036	3.11	0.47
SciDocs	scientific	25,657	1000	188	10.0	4.9	0.085	0.114	3.53	0.29
ArguAna	argument	8,674	1401	167	195.8	1.0	0.033	0.036	3.38	0.21
TREC-COVID	biomedical	171,332	50	180	11.7	493.5	0.151	0.092	2.65	0.45
Quora	general (dup. questions)	522,931	10000	12	9.6	1.6	0.051	0.044	4.92	0.74
Touche-2020	argument	382,545	49	316	6.7	19.0	0.042	0.008	3.26	0.75

Jargon = share of tokens rarer than Zipf 3 (about 1 per million words) in general English. Mean coverage = primary overlap measure. Five ArguAna queries whose relevant documents are missing from the corpus are excluded from overlap analyses. NFCorpus, SciFact and FiQA counts match the proposal's Table 10.

Validation against published references

BM25 reproduces the Pyserini BM25-flat reference numbers to within 0.005 nDCG@10 on all eight datasets, and our metric code matches pytrec_eval exactly on SciFact and FiQA (NFCorpus differs by 0.0005 because of tie ordering). See results/metric_validation.txt.

Dataset	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Ours (BM25, k1=0.9 b=0.4)	0.320	0.677	0.237	0.150	0.402	0.595	0.788	0.444
Pyserini BM25 flat (Kamalloo et al. 2024)	0.322	0.679	0.236	0.149	0.397	0.595	0.789	0.442

2. Results: effectiveness (E1, E2, E5, E6)

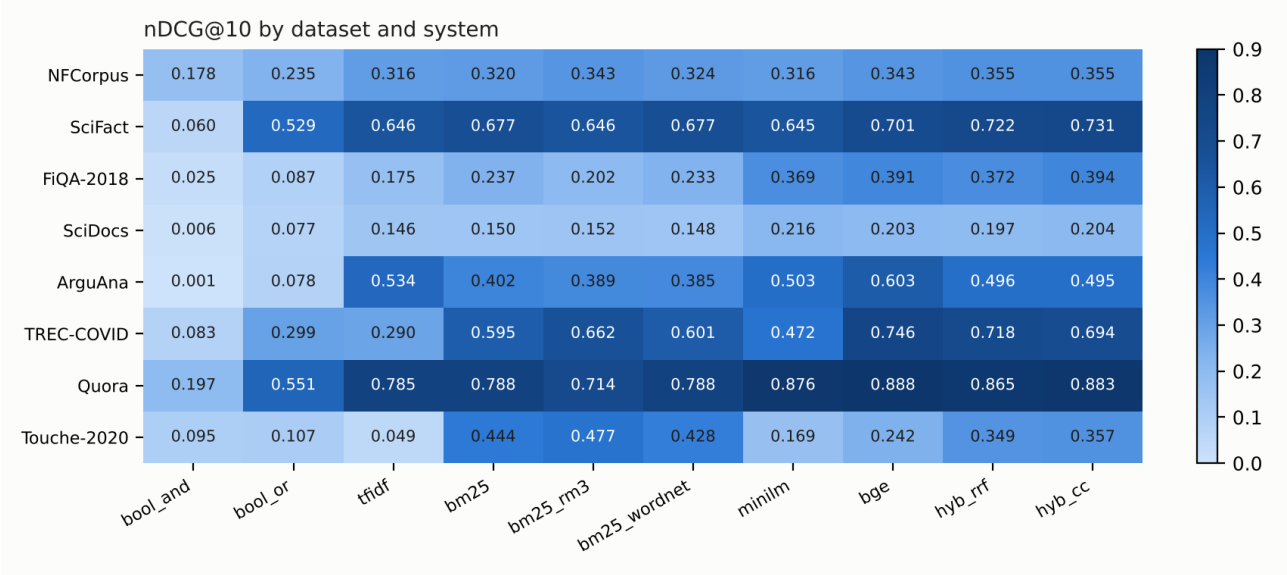


Figure 1. nDCG@10 for every system and dataset (darker = higher).

ndcg@10	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Boolean AND	0.178	0.060	0.025	0.006	0.001	0.083	**0.197**	0.095
Boolean OR (coord.)	0.235	0.529	0.087	0.077	0.078	0.299	**0.551**	0.107
TF-IDF / VSM	0.316	0.646	0.175	0.146	0.534	0.290	**0.785**	0.049
BM25	0.320	0.677	0.237	0.150	0.402	0.595	**0.788**	0.444
BM25 + RM3	0.343	0.646	0.202	0.152	0.389	0.662	**0.714**	0.477
BM25 + WordNet	0.324	0.677	0.233	0.148	0.385	0.601	**0.788**	0.428
MiniLM (dense)	0.316	0.645	0.369	0.216	0.503	0.472	**0.876**	0.169
bge-small (dense)	0.343	0.701	0.391	0.203	0.603	0.746	**0.888**	0.242
Hybrid RRF (MiniLM)	0.355	0.722	0.372	0.197	0.496	0.718	**0.865**	0.349
Hybrid convex (MiniLM)	0.355	0.731	0.394	0.204	0.495	0.694	**0.883**	0.357
Hybrid RRF (bge)	0.362	0.726	0.366	0.192	0.539	0.804	**0.868**	0.363
Hybrid convex (bge)	0.366	0.731	0.391	0.196	0.548	0.794	**0.887**	0.358

recall@100	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Boolean AND	0.106	0.059	0.033	0.006	0.001	0.008	**0.221**	0.194
Boolean OR (coord.)	0.211	0.800	0.280	0.219	0.392	0.055	**0.821**	0.307
TF-IDF / VSM	0.247	0.924	0.509	0.362	**0.979**	0.063	0.974	0.236
BM25	0.246	0.929	0.540	0.347	0.932	0.109	**0.973**	0.580
BM25 + RM3	0.310	0.909	0.514	0.366	0.956	0.133	**0.960**	0.593
BM25 + WordNet	0.249	0.920	0.531	0.346	0.921	0.106	**0.974**	0.578
MiniLM (dense)	0.311	0.925	0.706	0.510	0.977	0.085	**0.994**	0.417
bge-small (dense)	0.297	0.950	0.683	0.456	0.986	0.128	**0.995**	0.465
Hybrid RRF (MiniLM)	0.326	0.962	0.711	0.478	0.983	0.118	**0.996**	0.612

recall@100	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Hybrid convex (MiniLM)	0.323	0.958	0.716	0.484	0.984	0.122	**0.996**	0.607
Hybrid RRF (bge)	0.312	0.968	0.698	0.450	0.985	0.140	**0.996**	0.615
Hybrid convex (bge)	0.311	0.962	0.692	0.457	0.989	0.142	**0.995**	0.608

map@100	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Boolean AND	0.081	0.059	0.020	0.004	0.001	0.004	**0.187**	0.041
Boolean OR (coord.)	0.103	0.490	0.071	0.048	0.066	0.024	**0.509**	0.060
TF-IDF / VSM	0.143	0.589	0.142	0.099	0.454	0.033	**0.745**	0.032
BM25	0.145	0.638	0.192	0.101	0.332	0.074	**0.746**	0.264
BM25 + RM3	0.167	0.590	0.161	0.106	0.312	0.101	**0.663**	0.287
BM25 + WordNet	0.146	0.637	0.191	0.100	0.317	0.071	**0.746**	0.256
MiniLM (dense)	0.142	0.603	0.310	0.156	0.424	0.053	**0.843**	0.114
bge-small (dense)	0.155	0.662	0.334	0.142	0.530	0.094	**0.857**	0.157
Hybrid RRF (MiniLM)	0.173	0.676	0.309	0.140	0.417	0.087	**0.829**	0.243
Hybrid convex (MiniLM)	0.168	0.692	0.330	0.146	0.414	0.091	**0.851**	0.240
Hybrid RRF (bge)	0.176	0.684	0.305	0.136	0.457	0.109	**0.833**	0.244
Hybrid convex (bge)	0.177	0.692	0.331	0.138	0.463	0.112	**0.856**	0.239

mrr@10	NFCorpus	SciFact	FiQA-2018	SciDocs	ArguAna	TREC-COVID	Quora	Touche-2020
Boolean AND	**0.317**	0.062	0.038	0.015	0.001	0.196	0.205	0.261
Boolean OR (coord.)	0.405	0.494	0.110	0.150	0.057	**0.546**	0.543	0.275
TF-IDF / VSM	0.521	0.599	0.212	0.251	0.445	0.485	**0.780**	0.140
BM25	0.523	0.642	0.296	0.261	0.319	**0.856**	0.778	0.750
BM25 + RM3	0.523	0.599	0.248	0.254	0.298	**0.878**	0.692	0.786
BM25 + WordNet	0.524	0.642	0.289	0.258	0.304	**0.846**	0.778	0.751
MiniLM (dense)	0.505	0.605	0.445	0.360	0.413	0.724	**0.867**	0.320
bge-small (dense)	0.531	0.667	0.476	0.347	0.523	**0.942**	0.880	0.483
Hybrid RRF (MiniLM)	0.565	0.684	0.443	0.334	0.406	**0.931**	0.858	0.629
Hybrid convex (MiniLM)	0.563	0.698	0.466	0.344	0.403	**0.897**	0.877	0.624
Hybrid RRF (bge)	0.568	0.689	0.438	0.332	0.448	**0.987**	0.861	0.615
Hybrid convex (bge)	0.578	0.695	0.475	0.336	0.455	**0.990**	0.882	0.582

Bold = best system per dataset for that metric. Per-query values: results/perquery_*.csv.

Boolean retrieval as a set (its native output)

Strict conjunctive retrieval returns *nothing* for 40%-93% of queries on seven of eight datasets (Touche, with short keyword queries, is the exception at 6%). This is the vocabulary-mismatch problem in its most direct form: one missing term is enough to lose every relevant document.

Dataset	Empty AND result	Median set size	Set precision	Set recall	Set F1
NFCorpus	40%	1	0.253	0.108	0.123

Dataset	Empty AND result	Median set size	Set precision	Set recall	Set F1
SciFact	93%	0	0.060	0.059	0.059
FiQA-2018	68%	0	0.027	0.037	0.023
SciDocs	86%	0	0.013	0.006	0.007
ArguAna	92%	0	0.001	0.001	0.001
TREC-COVID	62%	0	0.141	0.012	0.019
Quora	69%	0	0.188	0.222	0.187
Touche-2020	6%	38	0.153	0.332	0.118

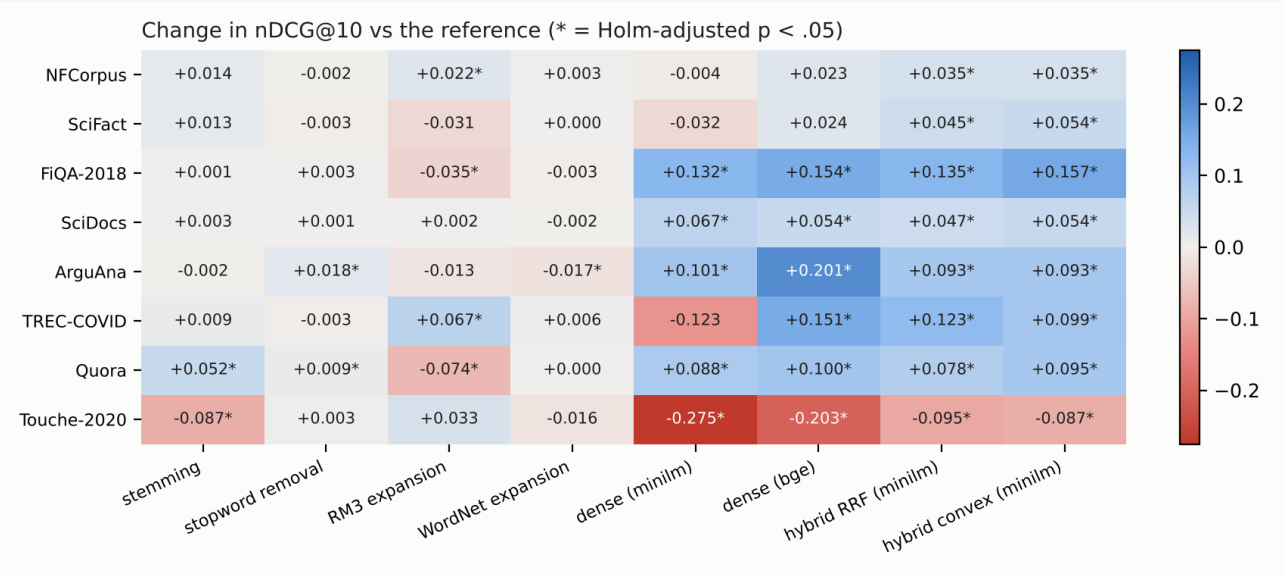


Figure 2. Effect of each intervention on nDCG@10. Preprocessing and expansion rows are relative to BM25 without that step / plain BM25; dense and hybrid are relative to BM25.

Stemming (E2) significantly helps only on Quora (+0.052) and significantly hurts on Touche (-0.087). Morphological variation is therefore a minor source of lexical failure on these collections. Stopword removal is negligible.

Query expansion (E5) does not close the vocabulary gap. WordNet expansion never significantly helps (0/8). RM3 significantly helps on 2/8 (NFCorpus, TREC-COVID) and significantly hurts on 2/8 (FiQA, Quora). Generic thesauri lack the domain-specific relations these queries need.

Hybrid retrieval is the most reliable intervention: convex BM25+MiniLM fusion significantly improves on BM25 in 7/8 datasets (+0.035 to +0.157) and is worse only on Touche. Dense-only MiniLM significantly beats BM25 on 4/8 and loses badly on Touche.

BM25 parameter sensitivity (oracle, tuned on test - upper bound only)

Dataset	BM25 default (0.9/0.4)	Oracle best	best k1	best b
NFCorpus	0.320	0.328	2.0	0.4
SciFact	0.677	0.690	2.0	0.9
FiQA-2018	0.237	0.253	1.5	0.75
SciDocs	0.150	0.160	2.0	0.9
ArguAna	0.410	0.529	2.0	0.9
TREC-COVID	0.595	0.658	2.0	0.6
Quora	0.790	0.814	0.9	0.75
Touche-2020	0.444	0.511	1.5	0.2

Tuning BM25 on test queries (an optimistic upper bound, computed on a 1,000-query sample for ArguAna and Quora, hence the slightly different defaults) gains up to 0.119 (ArguAna, b = 0.9). Stronger length normalisation recovers most of TF-IDF's ArguAna advantage, so H1's ArguAna reversal is a parameter effect, not a weakness of the BM25 model.

3. Hypothesis tests

H1. Is BM25 > TF-IDF > Boolean everywhere? Verdict: NOT SUPPORTED

BM25 beats both Boolean models everywhere, but it is **not** uniformly better than TF-IDF. The difference is not significant on NFCorpus, SciDocs and Quora; TF-IDF is significantly better on ArguAna, whose queries are whole arguments (about 196 tokens) where cosine length normalisation matters. On TREC-COVID and Touche, TF-IDF performs only at Boolean-OR level, because ltc weighting favours very short documents. The ordering is a property of model *and* collection, not a law. Full tests: results/h1_tests.csv.

H2. Does lexical retrieval degrade with domain specificity? Verdict: NOT SUPPORTED

Measured corpus jargon rate vs BM25 nDCG@10 across 8 domains: rho = 0.33 (p = 0.420); query jargon rho = -0.05; mean query IDF rho = 0.26. None is negative and significant. Raw nDCG is not comparable across datasets (different judging depth, number of relevant documents and task type), so the BM25-relative gain of a dense model is also shown. Its association with jargon is negative (dense helps *less* in jargon-heavy domains) but not significant with n = 8.

The proposal assumed specificity rises from NFCorpus to SciFact to FiQA. Measured jargon rates are FiQA 0.034 < NFCorpus 0.121 < SciFact 0.186. FiQA is the least specialised of the three.

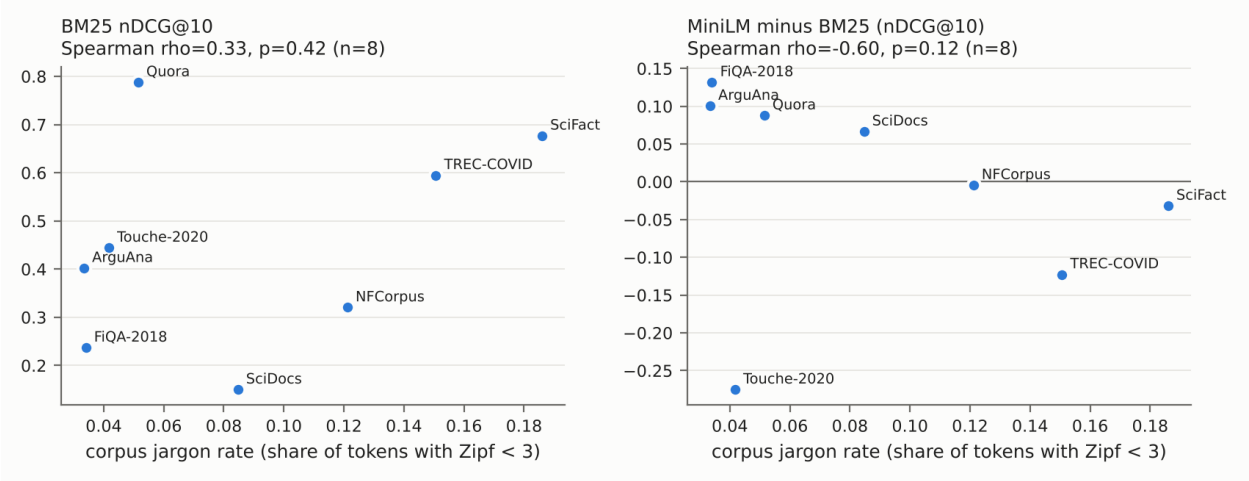


Figure 3. Domain-level jargon rate vs BM25 effectiveness (left) and vs the dense model's gain over BM25 (right). n = 8.

H3. Does query-level overlap explain BM25 effectiveness? Verdict: SUPPORTED

Within datasets, coverage strongly predicts BM25 nDCG@10 after controlling for query length and number of relevant documents (partial rho: NFCorpus 0.69, SciFact 0.73, FiQA-2018 0.49, SciDocs 0.51, ArguAna 0.66, TREC-COVID 0.37, Quora 0.44, Touche-2020 -0.03). The result holds for IDF-weighted coverage, maximum coverage and Jaccard (results/h3_correlations.csv). Touche is the exception: almost all its queries have high coverage, so there is no variance to explain.

Circularity check. The same coverage measure also predicts the dense model, whose scoring ignores lexical identity (partial rho for MiniLM: NFCorpus 0.53, SciFact 0.49, FiQA-2018 0.32, SciDocs 0.45, ArguAna 0.35, TREC-COVID 0.12, Quora 0.22, Touche-2020 -0.24). In the mixed model the coverage slope is 0.210 for BM25 vs 0.106 for MiniLM. Coverage is therefore partly a proxy for 'easy, well-specified query'. Only the difference between the two slopes is evidence of a lexical-specific mechanism.

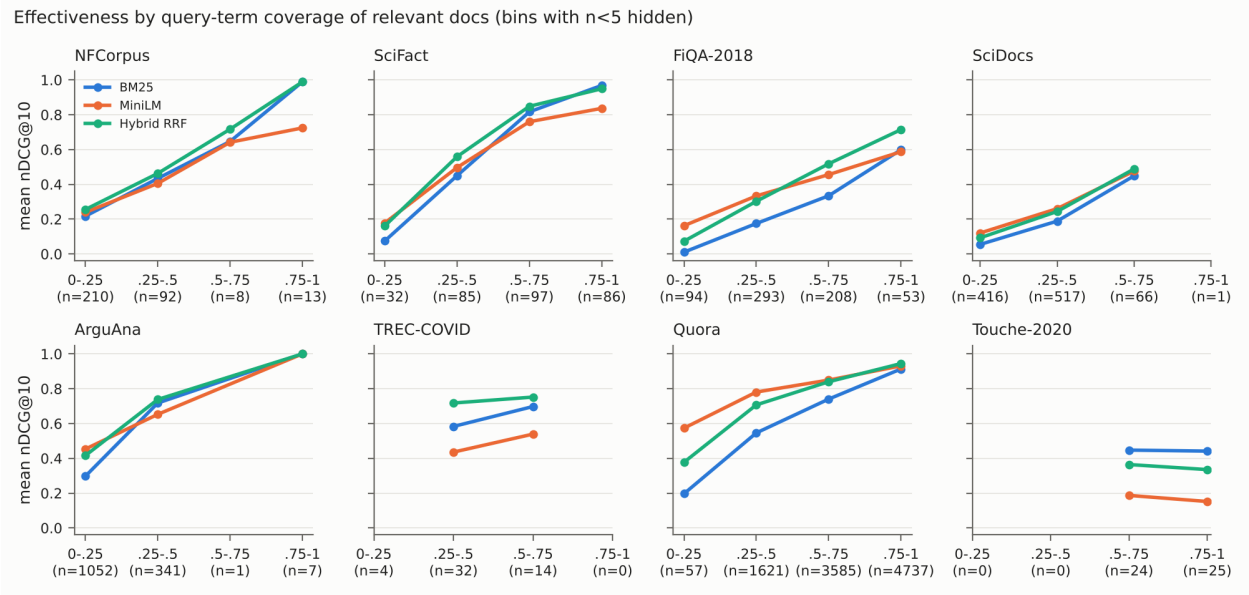


Figure 4. Mean nDCG@10 by coverage bin. The lines diverge at low coverage (dense and hybrid above BM25) on most datasets.

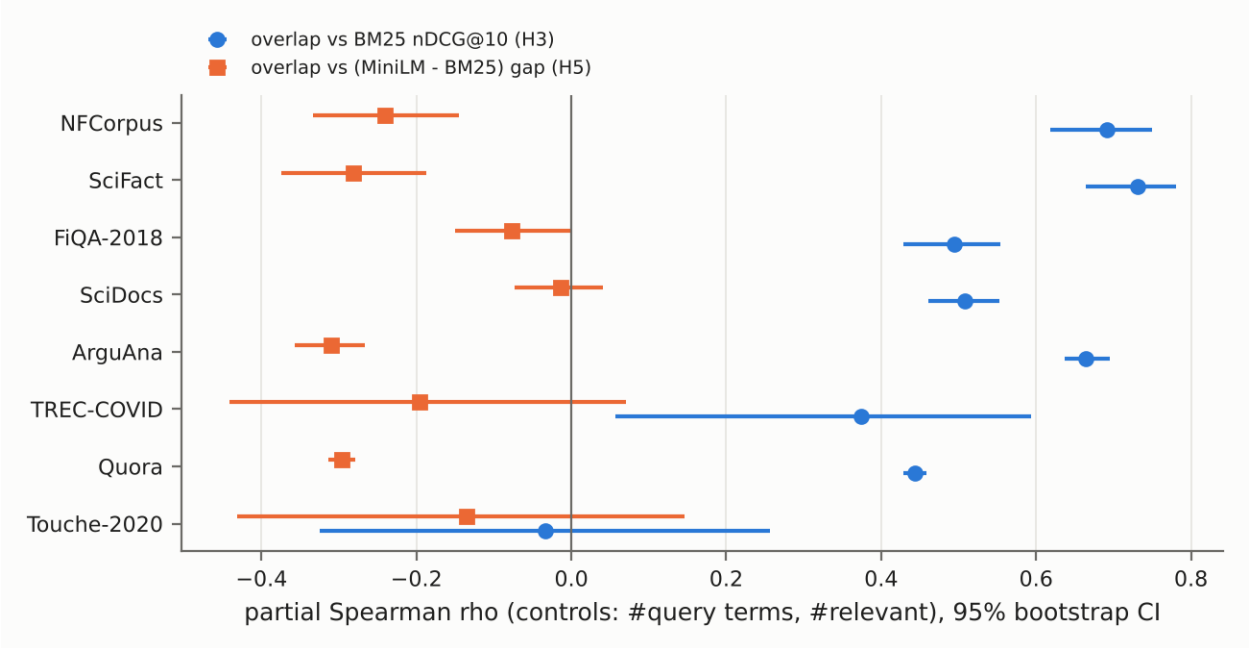


Figure 5. Partial correlations with 95% bootstrap CIs. Blue: coverage vs BM25 (H3). Orange: coverage vs the dense-minus-BM25 gap (H5).

H5. Does dense retrieval help most where overlap is low? Verdict: PARTIALLY SUPPORTED

Pooled over all queries, the dense model's advantage shrinks as coverage rises (mixed-model coefficient -0.104 nDCG per SD, $p < 0.001$; bge-small -0.108). Per dataset, the effect is significant on NFCorpus, SciFact, ArguAna and Quora but not on FiQA or SciDocs, where the dense model wins at *every* coverage level, nor on TREC-COVID or Touche (about 50 queries each). The decision rule needs significance in 6 of 8 datasets, so the verdict is partial support: the mechanism is real but its size depends on the collection.

H6. Is there a usable overlap threshold? Verdict: PARTIALLY SUPPORTED

A rule 'use a dense model when coverage < t' was learned on seven datasets and tested on the eighth. The threshold is stable (t = 0.60 in 7/8 folds), but held-out balanced accuracy averages only 0.56 (chance = 0.50) and the threshold-free AUC is 0.66. More fundamentally, coverage is computed from relevant documents, so it is an *offline diagnostic* for characterising a collection from a labelled sample, not a router for live queries. Domain-average coverage is also not significantly related to dense gains across the 8 domains.

H4. Failure-mode taxonomy Verdict: SUPPORTED (with caveats)

BM25 failures (nDCG@10 < 0.1) range from 2% of queries (TREC-COVID) to 54% (SciDocs). Up to 400 failures per dataset were labelled by rules (n = 1710). The distribution differs strongly by domain (Cramer's V = 0.24): NFCorpus, SciFact and ArguAna failures are dominated by paraphrase/indirect relevance (the relevant document shares almost no query terms), while FiQA, SciDocs and Quora have many term-competition failures (relevant documents match well, but so do many non-relevant ones, a polysemy/non-discriminative-term proxy). Abbreviation and synonymy, which the proposal emphasised, are minor.

Validity of the labels. Rules were revised once after a development audit (53% correct on 30 queries). On a fresh sample of 45 queries, 35 labels were judged correct (78%, 95% CI 64%-87%). Remaining errors are misspellings, clipped forms ('carb', 'corp') and occasional spurious WordNet senses. Audits: results/failure_manual_check_*.csv. Many NFCorpus 'failures' reflect how its relevance links were built (indirect topical links from a nutrition website) rather than lexical matching.

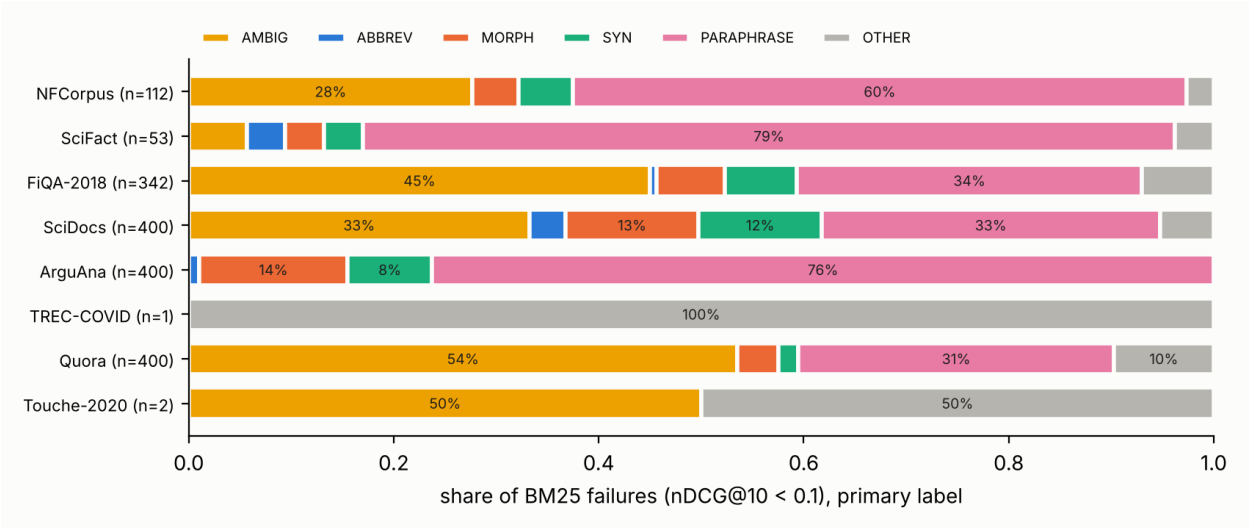


Figure 6. Primary failure mode per dataset (TREC-COVID and Touche have fewer than 3 failures and are excluded from the chi-square test).

4. Failure-mode catalogue (paraphrased examples)

Examples are paraphrased (not reproduced) as the proposal requires; mode labels follow the audited taxonomy. Two rows show modes the rules miss.

Mode	Domain	Paraphrased query	What the relevant document says instead
Paraphrase / indirect	Nutrition (NFCorpus)	A single root vegetable named as a query.	Bile-acid binding of steamed leafy greens and brassicas; the vegetable itself is never named.
Paraphrase / indirect	Nutrition (NFCorpus)	A brand-name cholesterol drug.	A trial of a fruit's effect on blood lipids in hyperlipidaemic patients.
Morphology	Nutrition (NFCorpus)	The noun for memory loss.	The adjective form inside the name of a shellfish-toxin poisoning syndrome.
Abbreviation	Science (SciDocs)	A paper title spelling out 'convolutional neural networks'.	Cited work refers only to 'CNN'.
Abbreviation / symbol	Science (SciFact)	A claim using a Greek-letter cytokine name and a disease acronym.	Abstract writes the Greek letter as a word and spells out the disease, so the tokens never match.
Synonymy	Science (SciFact)	A claim about deaths around birth caused by low birth weight.	A study of the global prevalence of childhood underweight.
Term competition	Finance (FiQA)	How inflation relates to interest rates.	The answer shares both key terms, but so do dozens of non-relevant posts that outrank it.
Term competition	General (Quora)	Best laptop for a given budget.	A near-duplicate question; many other budget-laptop questions match equally well.
Spelling variant	Science (SciDocs)	British spelling of a dialogue-systems term.	Cited paper uses the American spelling.
Paraphrase	Argument (ArguAna)	An argument that ideas cannot be owned, so drug patents are illegitimate.	The counter-argument talks about creative effort and generic production, with few shared words.
Misspelling (missed)	Science (SciFact)	A claim with a misspelt word for tumour spread.	Abstract uses the correct spelling.
Clipping (missed)	Nutrition (NFCorpus)	A diet named with a clipped word for carbohydrate.	Meta-analysis uses the full word.

Share of BM25 failures rescued (nDCG@10 >= 0.1) by each alternative

Primary mode	MiniLM (dense)	bge-small (dense)	Hybrid RRF (MiniLM)	BM25 + RM3	BM25 + WordNet
AMBIG	63%	64%	54%	7%	3%
ABBREV	45%	41%	36%	9%	5%
MORPH	49%	54%	33%	13%	3%
SYN	44%	40%	29%	5%	3%
PARAPHRASE	48%	54%	34%	11%	1%
OTHER	64%	66%	58%	7%	9%

Dense models rescue 44%-64% of BM25 failures in every mode, while WordNet expansion rescues at most 9% and RM3 at most 13%. Dense models are not specifically better on 'synonymy' failures, which is another sign that the lexical mode labels capture only part of what goes wrong.

5. Decision guideline

Based on these eight collections and two small dense models:

1. **Never use strict Boolean AND for ranked retrieval.** It returns no results for most queries.

2. **Default to a BM25 + dense hybrid** (convex fusion or RRF). It significantly improved on BM25 on 7 of 8 collections, by up to +0.16 nDCG@10, with a single loss on Touche. It also protects against choosing a weak dense model, as on TREC-COVID with MiniLM.

3. **BM25 alone is defensible when queries are short keyword queries over a collection whose relevance is topical and judged from lexical pools** (Touche: BM25 beats both dense models by 0.20-0.28), and, by a small margin, in jargon-heavy scientific/biomedical collections. Domain jargon is *not* a reason to abandon BM25.

4. **Tune BM25's length normalisation (b) on a development set before concluding lexical retrieval is weak.** Gains of up to 0.12 nDCG@10 were available (ArguAna).

5. **Do not expect generic query expansion to fix vocabulary mismatch.** WordNet did nothing and RM3 was mixed.

6. **Use coverage as an offline diagnostic.** Label about 50 queries; if most relevant documents contain fewer than about 60% of the query terms, a dense or hybrid component is likely to pay off. Treat this as a heuristic: it transferred

to unseen domains with only 0.56 balanced accuracy.

6. Threats to validity

• Eight domains give low power for domain-level tests; the jargon conclusions are 'no evidence for', not 'evidence against'.

• Both dense models are small (22-33M parameters); stronger models would likely widen the gaps, and results differ between the two.

• nDCG is not comparable across collections (judging depth, relevant documents per query, task). Within-dataset and relative measures carry the conclusions.

• Coverage is computed from relevance judgements, which in TREC-COVID and Touche were pooled mostly from lexical systems; this can favour BM25.

• Failure labels are heuristic (78% audited accuracy); the audit was done by one annotator (the analyst) without a second rater.

• The jargon measure (Zipf < 3) counts proper names and rare general words as jargon and ignores multi-word terms.

• BM25 grid search used test queries and is reported only as an upper bound. RM3 and WordNet parameters were not tuned.

• Dense runs were done on a different machine (Colab T4 GPU); a CPU cross-check on five datasets agreed to within 0.0004.

7. Reproducibility

Code: src/common.py (data, indexing, metrics), run_lexical.py (E1/E2/E5), run_dense.py (E6), features.py (E3), analysis.py (H1-H3, H5, H6, figures), failures.py (H4), validate_metrics.py, build_report.py; colab/ holds the GPU notebook. Order: run_lexical -> run_dense (or the Colab notebook) -> features -> analysis -> failures -> build_report.

All per-query results, test outputs and verdicts are in results/ (verdicts.json, h4_verdict.json, *_tests.csv, failure_manual_check_*.csv). Seed 42; BM25 k1 = 0.9, b = 0.4; dense max length 256.